

UEC521: DESIGN USING SYSTEM VERILOG

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Course Objective: The objective of this course is to provide methods and techniques to design and verify the functionality of digital electronic systems using System Verilog.

Syllabus

Data types: Built-in data types, fixed-size arrays, dynamic arrays, queues, associative arrays, array methods, creating new types with typedef, creating user-defined structures, enumerated types, constants, strings.

Design of Combinational and Sequential Circuits: HDL modelling of combinational and sequential circuits: multiplexer, decoder, adder, flip flops and registers, simple design examples, testbench for combinational and sequential circuits. Mealy and Moore FSMs, Design Examples.

Procedural statements and routines: procedural statements, tasks, functions, and void functions, routine arguments.

Connecting the test bench and design: Separating the test bench and design, the interface construct, stimulus timing, interface driving and sampling, program – module interactions, system Verilog assertions.

Basic OOP: Introduction, OOP terminology, creating objects, object deallocation, static variables vs. global variables, class methods, defining methods outside of the class, using one class inside another, understanding dynamic objects, copying objects, public vs. local, building a layered test bench.

Randomization: Randomization in System Verilog, Constraint, solution probabilities, controlling multiple constraint blocks, pre_randomize and post_randomize functions, random number functions, constraints tips and techniques, common randomization problems, iterative and array constraints.

Threads and interprocess communication: Threads, disabling threads, interprocess communication, events, semaphores, mailboxes and building a test bench with threads & interprocess communication (IPC).

Familiarization with standards: IEEE 1364-2005, IEEE 1800-2017.

Laboratory Work: Modelling and simulation of digital blocks, their verification by using System Verilog.

Course Learning Objectives (CLO)

The students will be able to:

1. Illustrate different data types used in System Verilog.
2. Design and test the functionality of digital system.
3. Apply the concepts of object oriented programming in System Verilog.
4. Apply randomization methods in System Verilog.

Text Books

1. Chris Spear, SystemVerilog for Verification A Guide to Learning the Testbench Language Features, Springer, (2008) 2nd Ed.
2. Stuart Sutherland, Simon Davidmann and Peter Flake, SystemVerilog for Design Second Edition: A Guide to Using SystemVerilog for Hardware Design and Modeling, Springer (2006); 2nd ed.

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